Fluid Mechanics ME 21201 Quiz 4

November 14, 2026

1. (1 pt) What is the physical meaning of the displacement thickness δ^* in an external boundary layer?

   A. The geometric thickness where the local velocity reaches exactly 99% of the free-stream velocity.

   B. The distance by which the external inviscid stream is displaced outward away from the body due to the mass deficit of the boundary layer.

   C. The vertical height above the plate where the viscous shear stress reaches its absolute maximum.

   D. The vertical coordinate where the local velocity equals the root-mean-square velocity of the flow.

   The distance by which the external inviscid stream is displaced outward away from the body due to the mass deficit of the boundary layer.

2. (1 pt) For steady, incompressible, laminar flow over a flat plate at zero angle of incidence, how does the local skin friction coefficient C_{fx} scale with the local Reynolds number Re_x?

   A. $C_{fx} \propto Re_x^{1/2}$

   B. $C_{fx} \propto Re_x^{-1}$

   C. $C_{fx} \propto Re_x^{-1/2}$

   D. $C_{fx} \propto Re_x^{-1/5}$

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3. (1 pt) In the classical Blasius similarity solution for laminar boundary layer flow over a flat plate, how does the hydrodynamic boundary layer thickness δ grow with the downstream distance x?

   A. $\delta \propto x^{1/5}$

   B. $\delta \propto x^{1}$

   C. $\delta \propto x^{3/4}$

   D. $\delta \propto x^{1/2}$

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4. (1 pt) What is the primary kinematic and dynamic condition at the wall that defines the exact point of boundary layer separation?

   A. The wall shear stress and the normal velocity gradient at the wall are both zero.

   B. The free-stream velocity drops to zero.

   C. The displacement thickness δ^* reaches a minimum.

   D. The local static pressure coefficient C_p is 0.

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5. (1 pt) Which of the following statements accurately describes the curvature/inflection point property of the Blasius velocity profile u/U_∞ across the boundary layer?

   A. It contains no inflection points anywhere in the domain $0 < y < \infty$.

   B. It possesses an inflection point ($\partial^2 u/\partial y^2 = 0$) right at the wall ($y = 0$).

   C. It has an inflection point exactly at the boundary layer edge ($y = \delta$).

   D. It has an inflection point at the displacement thickness height $y = \delta^*$.

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6. (1 pt) Why is a turbulent boundary layer generally more resistant to flow separation under an adverse pressure gradient than a laminar boundary layer?

A. Turbulent boundary layers have a much lower skin friction coefficient, which eliminates adverse pressure gradients.

B. Laminar boundary layers have higher kinetic energy in the inner sublayer than turbulent layers.

C. Turbulent viscosity completely eliminates static pressure variations in the boundary layer.

D. Turbulent eddies transport high-momentum fluid from the outer stream down toward the wall, energizing the near-wall region.

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7. (2 pts) In Thwaites' approximate integral method for boundary layers, one defines $\lambda = \frac{\theta^2}{\nu}\frac{dU}{dx}$, where U is the free stream velocity. Physically, what ratio does λ represent?

A. The ratio of the displacement thickness to the momentum thickness (δ^*/θ).

B. The ratio of turbulent kinetic energy production to viscous dissipation rate.

C. The ratio of convective inertial forces to viscous normal stresses in the free stream.

D. The ratio of pressure gradient forces to viscous shear/diffusion forces across the characteristic length scale θ.

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8. (1 pt) In what fundamental sense does the momentum integral equation satisfy Prandtl's boundary-layer PDE?

A. It satisfies Prandtl's differential equation in a weighted-average (global) sense across the transverse coordinate y from 0 to $\delta(x)$, rather than pointwise at every individual (x, y) coordinate.

B. It satisfies Prandtl's equation pointwise everywhere inside the boundary layer because an assumed polynomial velocity profile solves the Navier-Stokes equations exactly.

C. It replaces the viscous diffusion terms in the PDE with an inviscid Euler equation across the entire transverse profile.

D. It converts the PDE into an ordinary differential equation by treating pressure as a constant that is independent of x.

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9. (1 pt) Why do integral boundary layer methods consistently yield remarkably accurate estimates for skin friction C_f and momentum thickness $\theta(x)$ even when the assumed velocity profile shape (e.g., a simple cubic or quartic polynomial) differs noticeably from the exact Blasius profile?

A. The momentum integral equation completely eliminates velocity gradients and solves directly for surface pressure.

B. Any polynomial profile automatically satisfies the exact Navier-Stokes equations at every interior point of the boundary layer.

C. The integration over the transverse coordinate y acts as a spatial smoothing operator, making integral quantities (δ^*, θ) and skin friction relatively insensitive to local details of the velocity profile shape.

D. Skin friction on a flat plate depends exclusively on the free-stream velocity U_e and fluid viscosity ν, independent of the velocity profile near the wall.

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